Department of Mathematics & Applied Mathematics

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1 Introduction to the extra notes

These notes are in addition to the material in the main textbook. They have been written by a number of different lecturers over the years, so you may find that some of the notation changes from one place to another. Please keep an eye out for this!

Some of this will simply be additional material on top of a topic in Stewart, while some of it will cover an entire section which is not covered in Stewart. Please read through these and let me know if anything does not make sense. If that is the case, then I will do my best to add extra material.

If there is notation here which is not familiar, first see if you can figure it out from the context, then look it up online, then ask a classmate, then feel free to ask me and I will add some more explanation.

I will announce when a given section is needed, or your lecturer will let you know.

Here is a link to a useful video about writing mathematics.

2 Sets and functions

2.1 Sets

Informally we think of a set as a collection of objects (things). For a set A we say that a is an element of A when a is one of the objects in A and write $a \in A$; if a is not an element of A then we will write $a \notin A$. Sets are completely determined by their elements, meaning that two sets A and B are equal (written as $A = B$) if they have the same elements, i.e. each element of A is an element of B, and each element of B is an element of A. Sets can be finite (meaning that they contain finite number of elements), or infinite (meaning that they containing infinite number of elements). A set which has no elements, is called an empty set and is denoted usually by Ø.

Some important examples of infinite sets are:

- $\mathbb{N}$ - set of natural numbers
- $\mathbb{Z}$ - set of integers
- $\mathbb{Q}$ - set of rational numbers
- $\mathbb{R}$ - set of real numbers
- $\mathbb{C}$ - set of complex numbers (we will look at complex numbers in the second semester)

How can we describe a set?

List all the elements of that set (if we can) and enclose them in curly brackets, for example a set A whose elements are $1, 2, 3, 4$ can be written as $A = \{1, 2, 3, 4\}$. Note that the elements of a set are *unordered* and *distinct*. By unordered we mean that it does not matter in which order we list the elements, for example, the set $\{1, 2, 3, 4\}$ and $\{2, 1, 3, 4\}$ are the same; and by distinct we mean that we do not have two same elements in a set, for example the set $\{1, 2, 3, 4\}$ is the same (equal to) as the set $\{1, 1, 2, 3, 4\}$. If the elements of a set follow a pattern, then we can write, for example, the set of all even numbers from 2 to 20 as $A = \{2, 4, 6, ..., 20\}$, or in the case of an infinite set, for example, the set of all even numbers as $B = \{..., -2, 0, 2, 4, 6, ...\}$.

Now suppose we are given a set X and we know that some of its elements have the same "property". For example, if we take $X = \mathbb{R}$, then the elements of B above are divisible by 2. Then we can write B as the set $\{x \in \mathbb{R} | x = 2k, k \in \mathbb{Z}\}$, or $\{x \in \mathbb{R} : x = 2k, k \in \mathbb{Z}\}$, we could have also written $\{x \in \mathbb{Z} | x = 2k, k \in \mathbb{Z}\}$. In this notation, "|", or ":" means "such that" (for which), and we read it as "the set of all those x in $\mathbb{R}$ such that..". Now in general, if X is a set and we denote by $P(x)$ a property (statement) on x, then $\{x \in X | P(x)\}$ will be the set of all those x in X which have this property.

Subsets, union and intersection of sets

Definition 1. *A set S is said to be a* subset *of a set A, written as $S \subseteq A$, if each element of S is an element of A.*

As follows from this definition, to prove that a set S is a subset of a set A, we should take any element s in S and prove that it is also in A.

Note that S may equal to the set A. However, if S is a subset of A but does not equal to A, that is, every element of S is an element of A but there exists at least one element in A which is not in S, then we will use notation $S \subset A$; in this case, S is sometimes called a proper (or strict) subset of A. When S is not a subset of A we will write $S \not\subset A$.

Let's look at an example. Suppose $A = \{1,2,3\}$ and let us list all the subsets of A, they are: $S_1 = \emptyset$, $S_2 = \{1\}$, $S_3 = \{2\}$, $S_4 = \{3\}$, $S_5 = \{1,2\}$, $S_6 = \{1,3\}$, $S_7 = \{2,3\}$, $S_8 = \{1,2,3\}$. Now we can write $S_i \subset A$ when $i = 1,..,7$, and $S_i \subseteq A$ when $i = 1,...,8$. In general:

$S \subseteq A$ means that $S \subset A$ **or** $S = A$.

Going back to the examples above, we have the following inclusions:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

Note that any non-empty set A has always at least two subsets, they are A itself and Ø.

Exercise 2. *How many subsets does a set with n elements have?*

Definition 3. *Let X be a set and let A and B be subsets of X.*

- *The union of A and B, denoted by $A \cup B$, is the set which consists of all those elements which are in A **or** B. We can write this as:*

$$A \cup B = \{x \in X | x \in A \text{ or } x \in B\}$$

- *The intersection of A and B, denoted by $A \cap B$, is the set which consists of all those elements which are in both A **and** B. We can write this as:*

$$A \cap B = \{x \in X | x \in A \text{ and } x \in B\}$$

- *The set difference of A and B, denoted be $A \setminus B$, is the set of all those elements in A which are not in B. We can write this as:*

$$A \setminus B = \{x \in X | x \in A \text{ and } x \notin B\}$$

Two sets A and B are said to be disjoint if they do not have common elements, or equivalently their intersection is an empty set: $A \cap B = \emptyset$.

Exercise 4. *1. Let $A = \{a,b,c,d\}$, $B = \{b,d,f,h\}$, and $C = \{c,d,e,f\}$. Find:*
   - *(a) $A \cap B$, $A \cap C$, $B \cap C$;*
   - *(b) $A \cup B$, $A \cup C$, $B \cup C$;*
   - *(c) $A \setminus B$, $B \setminus A$, $B \setminus C$, $C \setminus B$, $A \setminus C$, $C \setminus A$.*

2. Given the set $\mathbb{R}$ of real numbers, let $A = \{x \in \mathbb{R}|\ 1 \le x \le 3\}$ and $B = \{x \in \mathbb{R}|\ 2 \le x \le 4\}$. Find:
   - *(a) $A \cap B$;*
   - *(b) $A \cup B$;*
   - *(c) $(\mathbb{R} \setminus A) \cap B$;*
   - *(d) $(\mathbb{R} \setminus B) \cap A$;*
   - *(e) $(\mathbb{R} \setminus A) \cap (\mathbb{R} \setminus B)$;*
   - *(f) $(\mathbb{R} \setminus B) \cup (\mathbb{R} \setminus A)$;*
   - *(g) $B \cup (A \cap (\mathbb{R} \setminus B))$;*
   - *(h) $((R \setminus A) \cap B) \cup ((\mathbb{R} \setminus B) \cap A)$.*

3. Let A, B, and C be any three sets. Prove the following:
   - *(a) $A \cap A = A$;*

(b) $A \cap B = B \cap A$;
(c) $A \cap (B \cap C) = (A \cap B) \cap C$;
(d) $A \cap B \subset A$, and $A \cap B \subset B$;
(e) $A \cap \emptyset = \emptyset$;
(f) $A \cup A = A$;
(g) $A \cup B = B \cup A$;
(h) $A \cup (B \cup C) = (A \cup B) \cup C$;
(i) $A \subset A \cup B$, and $B \subset A \cup B$;
(j) $A \cup \emptyset = A$;
(k) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
(l) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

4. *Let A and B be any two sets. Prove the following:*
   (a) $A \setminus A = \emptyset$;
   (b) $A \setminus \emptyset = A$;
   (c) $\emptyset \setminus A = \emptyset$;
   (d) $B \setminus A = \emptyset$ *if and only if* $B \subset A$;
   (d) $(A \setminus B) \cap (B \setminus A) = \emptyset$;
   (e) $A \cap (B \setminus A) = \emptyset$;
   (f) $A \setminus B = B \setminus A$ *is not always true.*
5. *Let $A = \{1, 0\}$, state whether each of the following statements are true or false:*
   (a) $\{1\} \in A$;
   (b) $\emptyset \in A$;
   (c) $\{1\} \subset A$;
   (d) $0 \in A$;
   (e) $1 \subset A$.
6. *Determine which of the following five sets are equal:* $\{a, b, c\}$, $\{b, c, a, b\}$, $\{c, a, c, b\}$, $\{b, c, b, a\}$, $\{a, b, c, d\}$.
7. *Let* $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7, 8, 9\}$, $C = \{2, 4, 6, 8\}$, $D = \{4, 5\}$, $E = \{5, 6\}$, $F = \{4, 6\}$, *and let X be a set which satisfies the following conditions:* $X \subset A$, $X \subset B$, *and* $X \not\subset C$. *Determine which of the sets A, B, C, D, E, F can equal to X.*
8. *Which of the following sets is the empty set?*
   (a) $\{x | x \text{ is an odd integer}\}$;
   (b) $\{x | x \text{ is an integer and } x + 8 = 8\}$;
   (c) $\{x | x \text{ is a negative integer and } x \geq 1\}$.
9. *Consider the following sets of figures in an Euclidean plane S:*

$$A = \{x | x \text{ is a quadrilateral in } S\},$$

$$B = \{x | x \text{ is a parallelogram in } S\},$$

$$C = \{x | x \text{ is a rhombus in } S\},$$

$$D = \{x | x \text{ is a rectangle in } S\},$$

$$E = \{x | x \text{ is a square in } S\}.$$

   (a) *Which sets are the subsets of the others?*
   (b) *Find the union and the intersection of each pair of sets.*

Intervals

For real numbers a and b such that $a < b$ the sets

$$\begin{aligned}
(a,b) &= \{x \,|\, a < x < b\} \\
[a,b] &= \{x \,|\, a \le x \le b\} \\
(a,b] &= \{x \,|\, a < x \le b\} \\
[a,b) &= \{x \,|\, a \le x < b\} \\
(a,\infty) &= \{x \,|\, a < x\} \\
[a,\infty) &= \{x \,|\, a \le x\} \\
(-\infty,b) &= \{x \,|\, x < b\} \\
(-\infty,b] &= \{x \,|\, x \le b\} \\
(-\infty,\infty) &= \mathbb{R}
\end{aligned}$$

are called intervals. Furthermore, the intervals (a,b), (a,∞), $(-\infty,b)$, and $(-\infty,\infty)$ are called open intervals, while the intervals $[a,b]$, $[a,\infty)$, $(-\infty,b]$, and $(-\infty,\infty)$ are called closed intervals.

Note that $-\infty$ and ∞ are not numbers!!

Inequalities

For real numbers a, b, c, and d, we have:

1. If $a < b$ and $b < c$ then $a < c$;
2. If $a < b$, then $a + c < b + c$;
3. If $a < b$ and $c < d$, then $a + c < b + d$;
4. If $a < b$ and $c > 0$, then $ac < bc$;
5. If $a < b$ and $c < 0$, then $ac > bc$;
6. If $0 < a < b$, then $\frac{1}{a} > \frac{1}{b}$.

Note that all exercises in blue are intended for enrichment

Exercise 5. *Prove each of the following:*

(a) Rule 3 follows from Rule 1 and 2;
(b) If $x < 0$, then $0 < -x$ (using Rule 2);
(c) Rule 5 follows from Rule 4 and (b);
(d) Rule 6 follows from Rule 4 and 1.

Exercise 6. *Solve the inequality*

(a) $1+x < 7x+5$ *We have:*

$$\begin{aligned}
& 1 + x < 7x + 5 \\
\Leftrightarrow & (1 + x) + (-1 - 7x) < (7x + 5) + (-1 - 7x) && \text{using Rule 2} \\
\Leftrightarrow & -6x < 4 \\
\Leftrightarrow & -6x(\frac{-1}{6}) > 4(\frac{-1}{6}) && \text{using Rule 4} \\
\Leftrightarrow & x > \frac{-2}{3}
\end{aligned}$$

(b) $1 - x > 2x + 1$.

Recall that $a \leq b$ is equivalent to the statement $a < b$ **or** $a = b$. It is not difficult to check that the rules of inequalities listed at the beginning of this section also hold if we replace $<$ by $\leq$ carefully i.e. we have:

1. If $a \leq b$, then $a + c \leq b + c$
2. If $a \leq b$ and $c \leq d$, then $a + c \leq b + d$;
3. If $a \leq b$ and $c \geq 0$, then $ac \leq bc$;
4. If $a \leq b$ and $c \leq 0$, then $ac \geq bc$;
5. If $0 < a \leq b$, then $\frac{1}{a} \geq \frac{1}{b}$.

Exercise 7. *Solve the inequalities*

(a) $2 \leq 4x - 5 \leq 11$;
(b) $x^2 - 7x + 12 \leq 0$;
(c) $x^3 - x^2 > 6x$.

2.2 Functions

Let us recall the definition of a function from the textbook: Let A and B be sets. A function f from A to B, written as $f : A \to B$, is a "rule" which assigns to each element x of A a unique (one and only one) element $f(x)$ in B. A is called the *domain* of f and B is called the codomain of f. The range of f is the set of all possible values of $f(x)$. That is, the range of f is the set $\{y \in B | \exists x \in A \text{ with } y = f(x)\}$, where the symbol $\exists$ denotes the word "exist". Note that the range of f is a subset of the codomain of f.

Remark 8. *Note that $f(x)$ is not a function! Here, f is a function while $f(x)$ is the value of f at x. When A and B are subsets of real numbers, then $f(x)$ is simply a number.*

Example 9 Let f be a function defined by $f(x) = x^2$. The domain of this function is R, while the range is $[0, \infty)$.

Example 10 Let f be a function defined by $f(x) = x$. The domain and the range of f is $\mathbb{R}$.

Exercise 11. *Find the domain of the function f defined by:*

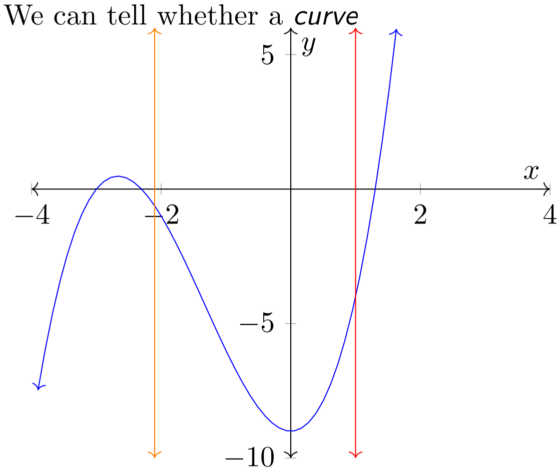
1. $f(x) = x + 1$;
2. $f(x) = \frac{1}{\sqrt{x+1}}$;
3. $f(x) = \frac{1}{x^2+2x-3}$.

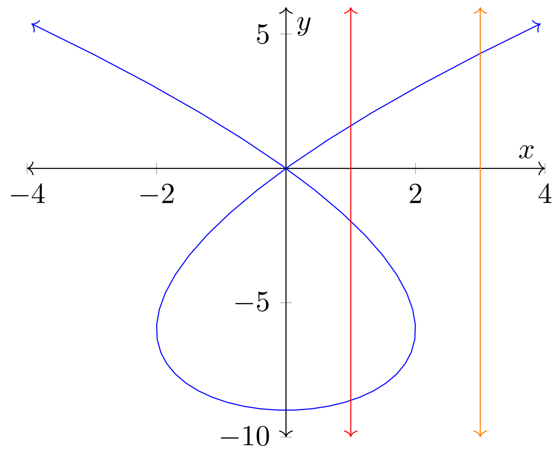
Exercise 12. *Find the range of the function f defined by:*

1. $f(x) = x + 1$;
2. $f(x) = \frac{1}{\sqrt{x+1}}$;
3. $f(x) = \frac{1}{x^2+2x-3}$;

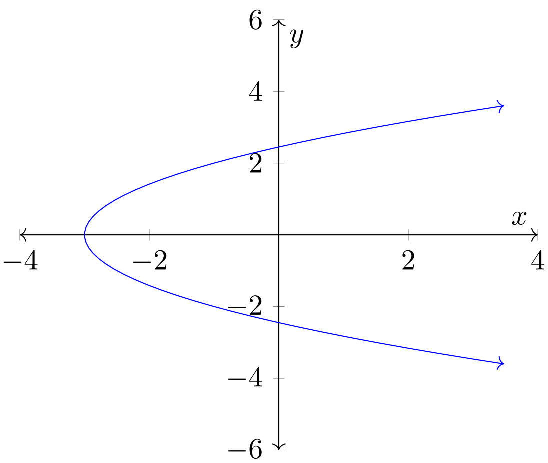
Let A and B be sets and let $f : A \to B$ be a function. The graph of f is a set of "ordered pairs": $\{(x, f(x)) | x \in A\}$. Denoting $y = f(x)$, the graph of f is the set $\{(x, y) | x \in A \text{ and } y = f(x)\}$.

We can tell whether a *curve*





is a function by checking that each line $x = c$ intersects the graph at at most one place. Note that this is sometimes called the vertical line test. Given a curve, for instance the one defined by $\frac{1}{2}y^2 - 3 - x = 0$



which is not a function, it is possible to find functions f and g defined in this case by

$$f(x) = \sqrt{2x + 6}$$
$$g(x) = -\sqrt{2x + 6}$$

whose graphs are contained in the graph of the given curve. Note that for the above curve x is a function of y.

Exercise 13. *Are the two functions f and g the only possible ones? If so prove that they are only these two. If not how many are there?*

Piece-wise defined functions

Sometimes it is convenient to describe a function f by specifying its values on various subsets of $\mathbb{R}$ (or more generally subsets of its domain). For example, we might define f to be the function defined by

$$f(x) = \begin{cases} x + 1 & \text{if } x > 5 \\ x^2 - 9 & \text{if } x \leq 5 \end{cases}$$

By this we mean that for $x \in \mathbb{R}$ the number $f(x)$ is defined by the formula $x + 1$ if $x > 5$ and by the formula $x^2 - 9$ if $x \leq 5$.

Exercise 14. *If f is defined as above, find the following values:*

1. $f(1)$;
2. $f(2)$;
3. $f(3)$;
4. $f(4)$;
5. $f(5)$;
6. $f(6)$;
7. $f(7)$;
8. $f(2\pi)$.

Now sketch the graph of f.

Absolute value

One of the examples of a piece-wise defined functions is the absolute value function. Informally we think of $|x|$, where $x \in \mathbb{R}$, as the distance from 0 to x (or equivalent from x to 0) on the number line. From this we would expect that $|x| \geq 0$ for all $a \in \mathbb{R}$, why? Formally we define

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0. \end{cases}$$

which means that $|x|$ is defined to be x when $x \geq 0$ and is defined to be $-x$ when $x < 0$. This determines a function $f : \mathbb{R} \to [0, \infty)$ defined as $f(x) = |x|$.

Exercise 15. *Find*

(a) $|1|$;
(b) $|-2|$ *Since $-2 < 0$ it follows by definition $|-2| = -(-2) = 2$;*
(c) $|3|$;
(d) $|-\pi|$.

Example 16 Express $|3x - 4|$ without using absolute value symbols.
According to the definition we have

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0. \end{cases}$$

and so setting $a = 3x - 4$ we find that

$$|3x-4| = \begin{cases} (3x-4) & \text{if } (3x-4) \geq 0 \\ -(3x-4) & \text{if } (3x-4) < 0. \end{cases}$$
$$= \begin{cases} (3x-4) & \text{if } x \geq \frac{4}{3} \\ -(3x-4) & \text{if } x < \frac{4}{3}. \end{cases}$$

Properties of absolute values

For all $a, b \in \mathbb{R}$ and $n \in \mathbb{Z}$:

1. $\sqrt{a^2} = |a|$;
2. $|ab| = |a||b|$, $b \neq 0$;
3. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$;
4. $|a^n| = |a|^n$;
5. If $a > 0$ then $|x| = a$ if and only if $x = a$ or $x = -a$;
6. $|x| < a$ if and only if $-a < x < a$;
7. $|x| > a$ if and only if $x > a$ or $x < -a$.

Exercise 17. *Solve*

(a) $|2x - 7| = 4$*;*
(b) $|x - 7| < 2$*;*
(c) $|5x - 3| \geq 2$*;*
(d) $|x - 4| < |x - 7|$*;*
(e) $|x^2 - 2x - 3| < 1$*;*
(f) $|x - 1| < |x - 2| + |x - 3|$.

Properties of absolute values, continued

1. $|a + b| \leq |a| + |b|$.

This inequality is called the triangle inequality (and holds in a more general context from which its name is derived). Here are two different proofs:

(i) We have

$$-|a| \leq a \leq |a|$$
$$-|b| \leq b \leq |b|$$

and hence adding these two identities we get

$$\begin{aligned} & -|a| + -|b| \leq a + b \leq |a| + |b| \\ \Leftrightarrow & -(|a| + |b|) \leq a + b \leq |a| + |b| \\ \Leftrightarrow & |a + b| \leq |a| + |b| \end{aligned}$$

where the last equivalent statement follows from Properties 2.2.4 and 2.2.5.

(ii) Since

$$|a + b|^2 = (a + b)^2 = a^2 + 2ab + b^2$$

and

$$(|a| + |b|)^2 = |a|^2 + 2|a||b| + |b|^2 = a^2 + 2|ab| + b^2$$

it follows that

$$(|a| + |b|)^2 - |a + b|^2 = 2|ab| - 2ab = 2(|ab| - ab)$$

and hence since $|ab| \geq ab$ we have that

$$(|a| + |b| - |a + b|)(|a| + |b| + |a + b|) = (|a| + |b|)^2 - |a + b|^2 \geq 0.$$

Therefore since $(|a| + |b| + |a + b|) > 0$ – unless $a = b = 0$ (in which case the identity is trivially true) – it follows that

$$|a| + |b| - |a + b| \geq 0$$

and hence that

$$|a + b| \leq |a| + |b|.$$

Exercise 18. *Given that*

$$|x-9| < 0.1 \text{ and } |y-6| < 0.2.$$

Use the triangle inequality to estimate

(a) $|(x+y)-15|$;
(b) $|(x+y)-16|$.

Exercise 19. *Using the triangle inequality show that if* $|x-5| < 0.1$ *then* $|x^2-4x-5| < 0.61$.

Logic and proofs 3

One of our aims in this course is to help you to read and understand proofs, and to write simple proofs of your own. To be able to do this, you will need to know a few things about elementary logic.

3.1 Basic logic

In what follows we introduce some basic ideas and terminology from logic.

Statements

- A (mathematical) *statement* or *proposition* is a sentence which is either true or false, but not both.
- When we *prove* something, we show that a *statement* is true.
- When we *disprove* something, we show that a *statement* is false.

Example 20. *(a) The sentence "91 is divisble by 7" is a proposition, since we can decide whether it is true or false (it is true).*
(b) The sentence "91 is a prime number" is a statement, since we can decide whether it is true or false (it is false).
(c) The sentence "This statement is false" is not a statement. (Check to see what happens if you assume it is false, and when it is true.)

Connectives

Two or more statements (propositions) can be linked by *connectives* to form a *compound statement* (*compound proposition*).

Example 21. *(a) "9 is an odd number" is a statement; so is "9 is a prime number". We can use the connective "and" to link the two statements, in this way creating the compound statement "9 is an odd number and 9 is a prime number" (This compound statement is false.)*
(b) We can also use the connective "or" to link statements. Using the two statements in (a) we can form the compound statement "9 is an odd number or 9 is a prime number". (This compound statement is true.)

The most commonly used connectives are the ones indicated by the words or phrases:

- and ($\wedge$)
- or ($\vee$)
- if ... then (implies) ($\Rightarrow$)
- if and only if ($\Leftrightarrow$)
- not ($\neg$).

(The symbols used as abbreviations for these connectives are given in brackets.)

Remarks:

(a) The connective "or" sometimes causes some confusion. The reason is that in everyday language we usually use the word "or" in an *exclusive* sense, that is, when we use it we mean either the one or the other, but not both. (For example, when we say "Today I am going to study mathematics or statistics", we mean that we are going to study the one, but not the other. But in mathematics "or" is used in the *inclusive* sense, that is in the sense of either the one, or the other, or both. This means, for example, that both the statements "2 is an even number or a prime number" and '3 is an even number or a prime number" are true.

(b) The word "not" is not strictly speaking a connective, since it is not used to link two statements, but rather to change the meaning of a statement by negating it. For example, "9 is a prime number" is a statement, and "9 is not a prime number" is a different statement obtained by using the word *not*. The second statement is called the *negation* of the first one.

(c) The precise meaning of these connectives can be given by making use of truthtables, but we will not do it here.

(d) For two statements P and Q, "if P then Q" means the same as "P implies Q".

(e) The compound statement "P if and only if Q" should be understood as **two** compound statements: "If P, then Q" **and** "If Q, then P". We'll see examples of this a little later.

Open sentences

Is the sentence "$x^2 > x$" a proposition?

You may have a problem with the fact that we are not told what x is. Let's avoid this problem for the moment by assuming that x is a real number. If we know this, can we say that "$x^2 > x$" is a statement? This depends on being able to decide whether it is true or false. Someone may say: "It is false; just take $x = \frac{1}{2}$." But then someone else may say: "It is true; just take $x = 2$." It now becomes clear that "$x^2 > x$" is sometimes true, and sometimes false; it all depends on the choice of x. As it stands, it does *not* qualify as a statement. We can think of x as being a *variable*. Whether the sentence is true or false will depend on the "value" this variable has; for some values it will be true, for others false. Here we use "value" for something that could be put in the place of a variable.

Similarly,the sentence

"If n is an odd number, then n^2 is an odd number"

looks like a compound proposition. The sentences

"n is an odd number"   and   "n^2 is an odd number."

are linked by the connective "if ... then". But are these sentences statements? Can we decide whether they are true or false? Surely that depends on the value of n. We again have sentences containing a variable; this time it can be taken to be an integer.

These two examples show that in mathematics we quite naturally come across sentences that contain variables, and that this disqualifies such sentences from being statements. You may quite rightly feel unhappy about not taking such sentences seriously. They seem to be saying something that makes sense. Perhaps we should have a second look at what they seem to be saying. But let's first agree on a name for such problematical sentences: we'll call a sentence containing variables an **o**pen sentence. Whether such a sentence is true or false depends on the values of the variables.

There is some more terminology associated with such open sentences. This is not essential to what follows, so if you are only interested in the basics, you may as well skip the next two paragraphs.

When we write down an open sentence, we usually have in mind a set of elements that can be substituted in the place of the variable(s) in the open sentence. We call this set the **u**niverse for the sentence. Ideally we should specify the universe for the open sentences we use; you will find that this is sometimes omitted if it is clear from the context. We have already suggested the set of real numbers as the universe for the open sentence "$x^2 > x$". For the open sentence "n is an odd number" the universe could be taken to be the set of integers.

Associated with an open sentence and a universe there is a **t**ruth set: the set of all the elements of the universe which when substituted in the open sentence makes it a true statement. So, for example, 2 is in the truth set of $x^2 > x$, but $\frac{1}{2}$ is not. The truth set in this case is the union of the intervals $(-\infty, -1)$ and $(1, \infty)$.

Quantifiers

But what is the use of open sentences if we cannot even decide whether they are true or false? We need something that will change an open sentence into a statement.

If we change the open sentence "$x^2 > x$" into

" For all real numbers x, $x^2 > x$"

we do get a statement, since we can decide whether it is true or false. (It is false, since we have an example of a real number x for which it fails.)

We can also change the open sentence "$x^2 > x$" into

" There is a real number x such that $x^2 > x$.

If we do this, we again get a statement, since we can decide whether it is true or false. (It is true.)

Phrases such as "for all" and "there is (are)" are called **q**uantifiers and they are used to **b**ind the variables in an open sentence to change it into a statement. The examples above introduce the two most important kind of quantifiers. Phrases such as *for all, for every, for any* are called **u**niversal quantifiers and denoted by the symbol $\forall$, while phrases such as *there is, there exists, for some* are called **e**xistential quantifiers and denoted by the symbol $\exists$.

The second example above is a little more difficult to analyse. You will probably agree that when we say

"If n is an odd number, then n^2 is an odd number"

we really mean

"It does not matter what the integer n is,
as long as n is an odd number, then n^2 is an odd number."

Another way to say this is:

"For all integers n, if n is an odd number, then n^2 is an odd number."

Here the quantifier "For all" is used to bind the variable n in the open sentence

"if n is an odd number, then n^2 is an odd number."

Once we have done this, it becomes a statement; we can in this case show it is true. We'll do so shortly.

Example 22. *The sentence*

"If n is a prime number, then $n+1$ is a prime number"

has a similar structure. It contains two sentences depending on the variable n, and linked by the connective "if ... then", and is itself an open sentence. The variable n can be taken to be a positive integer in this case. The open sentence can be changed into a statement by using the universal quantifier to get "For all natural numbers n, if n is a prime number, then $n+1$ is a prime number." It is a statement, because we can decide whether it is true or false. In this case it is false (why?). We could also use the existential quantifier: "There exists a natural number n such that if n is a prime number, then $n+1$ is a prime number." In this case we get a true statement (why?).

It is unfortunately true that we tend to be somewhat sloppy about inserting quantifiers when we state results. So, for example, you may well find that

"If n is an odd number, then n^2 is an odd number."

is stated as a result.

What is passed off as a statement when this is done is strictly speaking only an open sentence. The result, as we have seen, should read

"For every integer n, if n is an odd number, then n^2 is an odd number."

Such sloppiness is widespread, and you will come across it often in many mathematics texts you may read. It is assumed in such cases that it is clear what the quantifier is that is necessary to turn the open sentence into a true statement; almost invariably it will be some form of universal quantifier.

Example 23. *Suppose you are asked to prove or disprove the statement "The sum of two odd numbers is an even number." We can write this as "If n and m are odd numbers, then $n+m$ is an even number." This looks quite acceptable, but strictly speaking it is an open sentence (containing the variables n and m), rather than a statement. The intended statement is clearly: "For every integer n and every integer number m, if n is an odd number and m is an odd number, then $n+m$ is an even number." In this form it is clear that if we want to prove the statement, we will have to give an argument that is valid for every odd number n and every odd number m. An example of two odd numbers with a sum which is an even number will* ***not*** *prove the statement.*

You may feel that the insistence on putting in quantifiers is bordering on splitting hairs. What makes matters worse is that we are not going to adhere to our own strict standards in future. The reason that we are so petty now is that one should always be aware of the intended quantifier, even when it is not there explicitly. This becomes crucial when trying to prove that a statement is false, as we'll see later.

Example 24. *The connective "if and only if" (which is often abbreviated to "iff") is used to indicate that there are* ***t****wo implications. As an example, the sentence*

n is an odd number if and only if n^2 is an odd number

says two things:

- *If n is an odd number, then n^2 is an odd number,* ***and***
- *If n^2 is an odd number, then n is an odd number.*

Both of these are open sentences. What is intended here is the statement:

> *For every integer n, n is an odd number if and only if n^2 is an odd number.*

(Is this a true statement?)

When you are asked to prove a statement involving the connective "if and only if", keep in mind that there are two things you need to prove.

3.2 Definitions, proofs and counterexamples

Definitions

In some of the examples given above, the concepts *odd number, even number* and *prime number* ocurred. If we want to decide whether these statements are true or false, we will need to know exactly what these concepts mean. This means that we need **definitions** for these concepts. A definition gives us the meaning of something new in terms of things we know already. For example, we could define an even number as an integer which is divisible by 2. This definition assumes that we already know what an *integer* is, and also what *divisible by* means.

It is important when you use definitions to realise that a definition is a compound statement which uses the connective "if and only if". The confusing part here is that definitions are usually given with only an "if" where there should really have been an "if and only if".

Example 25. *The definition of an even number is often given as: An integer is an even number if it is divisible by 2. This should really read: An integer is an even number if and only if it is divisible by 2.*
The one implication enables us to test whether an integer is an even number. It says that if an integer is divisible by 2, it is even. We can therefore say 26 is an even number because it is divisible by 2.
The other implication helps us to prove things about even numbers. It says that if an integer is an even number, then it is divisble by 2. This enables us to prove, for example, that the square of an even number is also an even number. For if n is an even number, then it is divisible by 2, and therefore $n^2 = n \times n$ will also be divisible by 2.

Example 26. *The definition of a prime number is usually given as*

> *A natural number is a prime number if it is larger than 1 and divisible only by itself and 1.*

This should really be:

> *A natural number is a prime number if and only if it is larger than 1 and divisible only by itself and 1.*

As an exercise, use the definition to show that 7 is a prime number, and also to prove that the square of a prime number is not a prime number.

When reading a definition, keep in mind that even if it only says "if", it really means "if and only if".

Implications and converses

Most of the theorems you'll see in this course (and elsewhere) will be statements of the form "If $P(x)$, then $Q(x)$", where $P(x)$ and $Q(x)$ are open sentences depending on the variable x. As we have seen, what is really meant by this is "For every x, if $P(x)$, then $Q(x)$" To prove that such a statement is true, we assume that the statement $P(x)$ is true, and show that it follows from this that the statement $Q(x)$ is true, making sure that our argument is valid for every x in the universe for the open sentences $P(x)$ and $Q(x)$. The variable x need not be a number; it could, for example, be a function.

Example 27. *We prove the statement "Every increasing function is one-to-one". To be more precise, we prove "For every function $f : \mathbb{R} \to \mathbb{R}$, if f is increasing, then f is one-to-one". (The notation $f : \mathbb{R} \to \mathbb{R}$ means f is a function from the real numbers to the real numbers.) We need two definitions:*

- *A function $f : \mathbb{R} \to \mathbb{R}$ is increasing if $x_1 < x_2$ implies that $f(x_1) < f(x_2)$.*
- *A function f is one-to-one if $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.*

Now suppose f is an increasing function, and let $x_1 \neq x_2$. Then either $x_1 < x_2$ or $x_2 < x_1$. If $x_1 < x_2$ then it follows from the fact that f is increasing that $f(x_1) < f(x_2)$, and so $f(x_1) \neq f(x_2)$. If $x_2 < x_1$, then since f is increasing, $f(x_2) < f(x_1)$ and so $f(x_1) \neq f(x_2)$. This proves that if $x_1 \neq x_2$, the $f(x_1) \neq f(x_2)$, and so f is one-to-one. Since the proof works for every function $f : \mathbb{R} \to \mathbb{R}$, we have proved the statement.

When working with implications, the *order* in which we write down things is very important.

Look at the statements:

(a) For every function $f : \mathbb{R} \to \mathbb{R}$, if f is increasing, then f is one-to-one.

(b) For every function $f : \mathbb{R} \to \mathbb{R}$, if f is one-to-one, then f is increasing.

These two statements clearly do **not** say the same thing. We have proved above that statement (a) is true. However, we **c**annot assume that because statement (a) is true, statement (b) will also be true. (It is false, in fact. Why?).

More generally, if P and Q are statements, then $P \Rightarrow Q$ and $Q \Rightarrow P$ are both statements, but different ones. We call the statement $Q \Rightarrow P$ the **c**onverse of the statement $P \Rightarrow Q$.

WARNING:

$$\boxed{\text{If a statement of the form } P \Rightarrow Q \text{ is true, its converse } Q \Rightarrow P \text{ need not be true.}}$$

We tend to use the term "converse" rather loosely. Even though "if n is an odd number, then n^2 is an odd number" is an open sentence rather than a statement, we'll refer to the open sentence "if n^2 is an odd number, then n is an odd number" as its converse. We'll also call the statement (b) above the converse of the statement (a).

Counterexamples

We have claimed above that the statement

$$\text{For every function } f : \mathbb{R} \to \mathbb{R}, \text{ if } f \text{ is one-to-one, then } f \text{ is increasing} \qquad (*)$$

is *not* true. How do we prove that a statement of this form is false? The statement claims that *for every* function something is true. To show that the claim is false, it is enough to give an example of *one* function for which that something is not true. The "something" in this case is the implication "if f is one-to-one, then f is increasing". This means that to disprove the statement $(*)$ (that is, to show that it is false) we have to find one example of a one-to-one function f such that f is not increasing. The function f defined by $f(x) = -x$ is such a function.

An example like this that is used to disprove a statement containing a universal quantifier is called a **c**ounterexample.

Proof by contradiction

We showed earlier that the square of an even number is an even number. More precisely, we proved that for every integer n, if n is even, then n^2 is even. Is the converse of this statement also true? The converse is

For all integers n, if n^2 is an even number, then n is an even number.

If you look at some examples, it seems pausible that this statement may well be true. (For example, $2^2 = 4$ and $4^2 = 16$ are even, and 2 and 4 are even.) But evidence in the form of examples is **not** a proof. We have to give an argument that will show that the statement is true for **all** integers.

A direct proof would start by assuming that n is an integer such that n^2 is an even number. This means that n^2 is divisible by 2, and so we can find some integer k such that $n^2 = 2k$. But where does one go from here? When it is not obvious how to do a direct proof, it is often a good idea to try a **proof by contradiction**. We show how this method works in this case.

Suppose that n is an integer such that n^2 is an even number, but that n is not an even number, that is, it is an odd number. Then n is not divisible by 2, and it must therefore leave a remainder when we divide by 2. The only possible non-zero remainder when dividing by 2 is 1. This means that we must be able to write the odd number n in the form $n = 2m + 1$, for some integer m. Then $n^2 = (2m+1)^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2) + 1$, and this shows that n^2 leaves a remainder 1 when we divide by 2, and so n^2 is not even. But this contradicts the fact that n^2 is even, and therefore our assumption that n is not even was wrong, and it must be even. (We have shown that the assumption that n^2 is even *and* n is odd leads to a contradiction, and therefore cannot be true. But if it is not possible for n to be odd when n^2 is even, then if n^2 is even, n must be even.)

In general, to prove a statement of the form $P \Rightarrow Q$ using the method of proof by contradiction, we assume that the statement P is true, and suppose that Q is false. Using these two assumptions, we argue until we get a statement that is clearly false (this is the *contradiction*). The only way to explain this contradiction is that the assumption that P is true and Q is false was wrong. Hence if P is true, Q must be true as well.

Contrapositives

There is a closely related method of proof for statement of the form $P \Rightarrow Q$ that uses the concept of the *contrapositive* of a statement. First recall that $\neg Q$ is shorthand for "not Q". A statement of the form

$$P \Rightarrow Q$$

is *logically equivalent* to the statement

$$\neg Q \Rightarrow \neg P$$

(one can show this using truth tables). We write this as

$$P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P.$$

This means that if we can prove that one of the statements $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ is true, the other will be true as well. We call $\neg Q \Rightarrow \neg P$ the **c**ontrapositive of $P \Rightarrow Q$. There are cases where it turns out to be easier to prove the contrapositive than the statement itself.

We have not looked at what is meant by the contrapositive of a statement that contains a quantifier. We'll say that the contrapositive of a statement of the form

$$\text{For every } x,\ P(x) \Rightarrow Q(x)$$

is

$$\text{For every } x,\ \neg Q(x) \Rightarrow \neg P(x) \qquad .$$

Example 28. *Let us consider again the statement*

> *For every integer n, if n^2 is an even number, then n is an even number.*

The contrapositive of this statement is

> *For every integer n, if n is not an even number, then n^2 is not an even number.*

or, equivalently,

> *For every integer n, if n is an odd number, then n^2 is an odd number.*

We leave it to you to prove this statement; the method is very similar to the one in the proof by contradiction above.

Polynomials 4

4.1 Introduction

An expression of the form

$$a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1} + a_nx^n,$$

where n is a nonnegative integer, $a_0, a_1, a_2, \ldots, a_n$ are real numbers, and $a_n \neq 0$, is called a *polynomial* in the variable x and the number n is the *degree* of the polynomial, i.e., n is the highest power of x with a non-zero coefficient[1].

Example 29 The function $3 + 2x - \frac{2}{5}x^{11}$ is a polynomial of degree 11. Here, $a_0 = 3$, $a_1 = 2$, $a_{11} = -\frac{2}{5}$, and for all $k \notin \{0, 1, 11\}$, we have $a_k = 0$. This simply says that for all the coefficients a_k where k is not 0,1 or 11, the value of a_k is 0.

Given a polynomial $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, we can define a function $p : \mathbb{R} \to \mathbb{R}$ (it takes in a real number and returns a real number) by

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n.$$

This is called a *polynomial function*. In future, we'll refer to these simply as polynomials.

Example 30 If $a_1 \neq 0$, then $p(x) = a_0 + a_1x$ is a polynomial function of degree 1, or simply a polynomial of degree 1. A polynomial of degree 1 is also called a *linear function*.

Example 31 If $a_2 \neq 0$, then $a_0 + a_1x + a_2x^2$ is a polynomial of degree 2, which is also called a *quadratic function*.

A polynomial of degree 3 is also called a *cubic function*, at degrees 4 and 5 we have *quartic* and *quintic* functions respectively.

If only a_0 is non-zero then we simply have a constant function.

4.2 Zeros of polynomials

A number $a \in \mathbb{R}$ is called a *zero* or *root* of the polynomial $p(x)$ if $p(a) = 0$.

Example 32 If $p(x) = a_0 + a_1x + a_2x^2$, then the zeros of $p(x)$ are

$$x = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0a_2}}{2a_2}$$

provided that

$$\Delta = a_1^2 - 4a_0a_2 \geq 0.$$

[1]Using sigma notation, a polynomial of degree n can also be written as $\sum_{i=0}^{n} a_ix^i$. If you don't know about sigma notation, have a look at the link:Khan Academy Sigma Notation.

The number $\Delta = a_1^2 - 4a_0a_2$ is called the *discriminant* of $p(x)$. The discriminant plays an important role in the study of polynomials:

- If $\Delta < 0$, then $p(x)$ has no real zeros[2].
- If $\Delta = 0$, then $p(x)$ has exactly one real zero.
- If $\Delta > 0$, then $p(x)$ has exactly two real zeros.

Example 33 Suppose $p(x) = 2x^2 - 3x + 1$. Then $\Delta = 9 - (4 \times 2 \times 1) = 1$, so $p(x)$ has two real zeros, namely

$$x_1 = \frac{3 + \sqrt{1}}{4} = 1$$

and

$$x_2 = \frac{3 - \sqrt{1}}{4} = \frac{1}{2}.$$

4.3 Polynomial division

Let's start by reminding ourselves how we divide integers.

Example 34 $47 \div 5$ gives 9 with a remainder of 2. We can write this as

$$47 = \underbrace{9}_{\text{quotient}} \times \underbrace{5}_{\text{divisor}} \quad + \quad \underbrace{2}_{\text{remainder}}$$

and note that $0 \leq 2 < 5$, i.e., $0 \leq$ remainder $<$ divisor.

More generally, for any pair n, d of positive integers, we can find nonnegative integers q and r, where $r < d$, such that

$$n = qd + r.$$

Something similar holds for polynomials.

Division Theorem for Polynomials:

Take a look here for some useful online materials if you are still not sure about the below: Khan Academy videos (https://www.khanacademy.org/math/algebra2/arithmetic-with-polynomials/long-di of-polynomials/v/polynomial-division).

If $p(x)$ and $d(x)$ are polynomials, then there are polynomials $q(x)$ and $r(x)$, where degree of $r(x) <$ degree of $d(x)$, such that

$$p(x) = q(x)d(x) + r(x).$$

[2]Why we say *real* zeros will become clear later in the course when we discuss complex numbers.

Example 35 Suppose we want to divide $p(x) = 3x^4 - 2x^3 + 4x - 5$ by $d(x) = x^2 + x$. Then:

$$
\begin{array}{r|llllll}
 & 3x^2 - 5x + 5 & & & & & \\
\cline{2-7}
x^2 + x & 3x^4 & -\,2x^3 & +\,0x^2 & +\,4x & -\,5 \\
 & \underline{3x^4} & \underline{+\,3x^3} & & & \\
 & & -\,5x^3 & +\,0x^2 & & \\
 & & \underline{-\,5x^3} & \underline{-\,5x^2} & & \\
 & & & 5x^2 & +\,4x & \\
 & & & \underline{5x^2} & \underline{+\,5x} & \\
 & & & & -\;\;x & -\,5
\end{array}
$$

Thus

$$
\begin{array}{ccccc}
3x^4 - 2x^3 + 4x - 5 & = & (3x^2 + 5x - 5) & (x^2 + x) & + (-x - 5) \\
p(x) & = & q(x) & d(x) & + \quad r(x)
\end{array}
$$

If you can't follow the above, then write down a long division of integers and see what steps you are taking, and then work out what the analogous steps would be for polynomials.

A special case of the Division Theorem for Polynomials occurs when $d(x) = x - a$. Here, the degree of $d(x) = x - a$ is 1. Since degree of $r(x) <$ degree of $d(x)$, it follows that $r(x)$ is a constant, C. Writing

$$p(x) = q(x)(x - a) + r(x)$$

and setting $x = a$, we obtain

$$p(a) = (q(a) \times 0) + r(a).$$

Hence, $r(x) = C = p(a)$.

Example 36 Suppose that we wish to divide $p(x) = x^3 - 2$ by $d(x) = x + 1$. We find that $x^3 - 2 = (x^2 - x + 1)(x + 1) - 3$. Notice that $r(x) = -3 = p(-1)$.

Notice the following:

- Suppose that a is a zero of $p(x)$. Then $C = p(a) = 0$, which means that $p(x) = q(x)(x - a)$, i.e., $x - a$ is a factor of $p(x)$.

- If, on the other hand, $x-a$ is a factor of $p(x)$, then $r(x)=0$. Setting $x=a$, we then get

$$p(a)=q(a)(a-a)+0=0,$$

which means that a is a zero of $p(x)$.

Putting these two observations together, we get the next result, which is useful when factoring polynomials.

Factor Theorem for Polynomials:

Let $p(x)$ be a polynomial. Then $p(a)=0$ if and only if $x-a$ is a factor of $p(x)$ (ie. $x-a$ divides $p(x)$ with no remainder).

Example 37 Suppose we want to factor the polynomial

$$f(x)=x^3-x^2-x+1.$$

We notice that $f(1)=0$. By the Factor Theorem, it follows that $x-1$ is a factor of x^3-x^2-x+1. Then:

$$
\begin{array}{r|llll}
 & x^2-1 & & & \\
\hline
x-1 & x^3 & -x^2 & -x & +1 \\
 & x^3 & -x^2 & & \\
\cline{2-3}
 & & 0 & -x & +1 \\
 & & & -x & +1 \\
\cline{3-5}
 & & & & 0
\end{array}
$$

Thus

$$
\begin{aligned}
x^3-x^2-x+1 &= (x^2-1)(x-1)=(x+1)(x-1)(x-1) \\
&= (x-1)^2(x+1).
\end{aligned}
$$

Alternatively, we can do the following:

$$
\begin{aligned}
f(x) &= x^3-x^2-x+1 \\
&= x^2(x-1)-(x-1) \\
&= (x^2-1)(x-1) \\
&= (x-1)^2(x+1).
\end{aligned}
$$

4.4 Factoring $1 - x^n$

Suppose we want to find a formula for the sum

$$S = 1 + x + x^2 + x^3 + \cdots + x^{n-2} + x^{n-1},$$

where n is some positive integer. Notice that

$$xS = x(1 + x + x^2 + x^3 + \cdots + x^{n-2} + x^{n-1}) = x + x^2 + x^3 + x^4 + \cdots + x^{n-1} + x^n.$$

Thus,

$$\begin{aligned} S - xS &= \left(1 + x + x^2 + x^3 + \cdots + x^{n-2} + x^{n-1}\right) - \left(x + x^2 + x^3 + x^4 + \cdots + x^{n-1} + x^n\right) \\ &= 1 + (x - x) + (x^2 - x^2) + (x^3 - x^3) + \cdots + (x^{n-1} - x^{n-1}) - x^n \\ &= 1 - x^n \end{aligned}$$

But $S - xS = S(1 - x)$, so when $x \neq 1$ we have

$$S = \frac{1 - x^n}{1 - x}.$$

Another way of writing this, which is true for all x, is

$$1 - x^n = (1 - x)(1 + x + x^2 + \cdots + x^{n-1}).$$

Example 38 We can write

$$1 - x^4 = (1 - x)(1 + x + x^2 + x^3).$$

4.5 The behaviour of polynomials for large positive or negative values of x

Example 39 How does the polynomial

$$f(x) = x^3 - x^2 - x + 1$$

behave when x is a large positive or large negative number? Factoring out the highest power of x from $f(x)$, we find

$$f(x) = x^3\left(1 - \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3}\right).$$

For large positive or negative values of x, the terms $1/x$, $1/x^2$, and $1/x^3$ are all very close to 0. On the other hand, the constant term 1 in the brackets on the right hand side does not change. So for large positive or negative values of x, we can write

$$\begin{aligned} f(x) &\approx x^3\,(1 + 0 + 0 + 0) \\ &= x^3 \end{aligned}$$

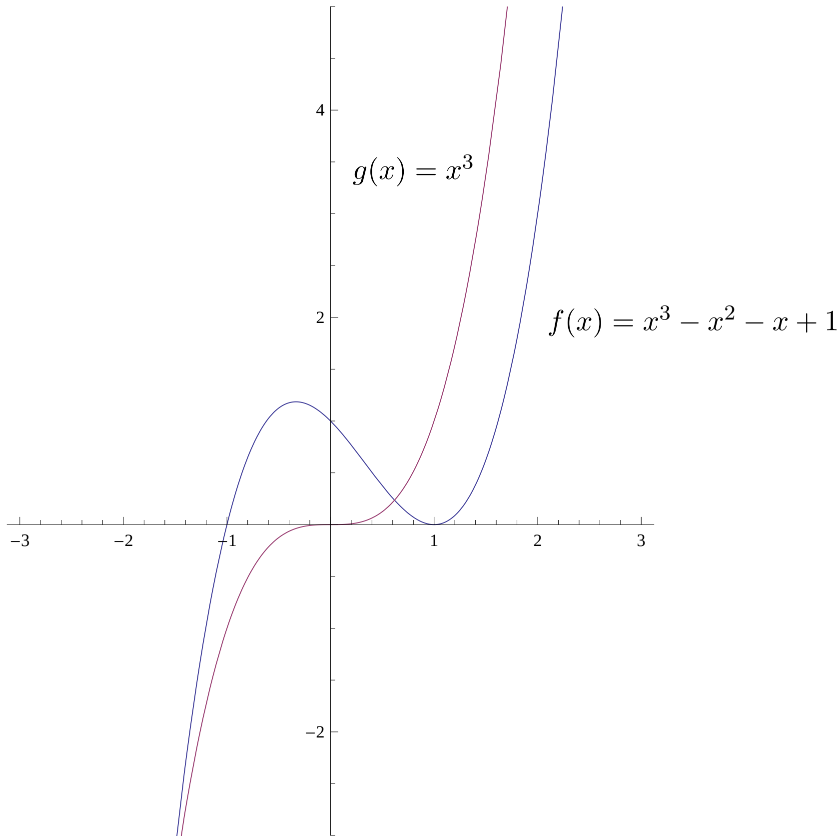


Figure 4.1: The two functions $f(x) = x^3 - x^2 - x + 1$ and $g(x) = x^3$.

Thus, the behaviour of $f(x)$ for large positive or negative values of x is determined by the behaviour of the term with the highest power of x. This is illustrated in Figure 4.1, where $f(x) = x^3 - x^2 - x + 1$ and $g(x) = x^3$ are shown on the same set of axes.

5 Mathematical induction

5.1 Introduction

In this section we introduce the principle of Mathematical Induction. This is a method of proving mathematical statements, and is extremely powerful. Often it can be used to effectively prove an infinite number of statements in one go, through a relatively simple procedure.

First however, we need to look at what we mean by a mathematical statement. Take the following definitions of positive, and nonnegative:

A real number x is

- *positive* if $x > 0$,
- *nonnegative* if $x \geq 0$.

Statements:

- The number 2 is both positive and nonnegative.
- The number 0 is nonnegative but not positive.
- The smallest positive integer is 1,
- The smallest nonnegative integer is 0.

A *statement* is something that is true or false. Here are a few more examples of slightly less trivial statements:

Example 40

- 'If you square any odd number and divide the answer by 4, the remainder is 1' is a statement (a true one, as you proved in Extra Problems 1).
- 'All cars are red' is a statement (a false one).
- 'If $(x-1)(y-1) = 1$, then $x = 2$ and $y = 2$' is a statement (a false one, see Extra Problems 1 again).
- '$x = 2$ is a solution of the equation $2x - 4 = 0$' is a statement (a true one).
- 'If $x^2 = 9$, then $x = 3$' is a statement (a false one, since it's possible that $x = -3$).
- '$x^2 - 5x + 7$' is not a statement (because it's not true or false) but $x^2 - 5x + 7 = 16$ is a statement. Whether it's true or false depends on the value of x.
- '$3^n + 1$ is divisible by 4' is a statement that is true for some values of n (like $n = 1$) and false for other values of n (like $n = 2$). On the other hand, '$3^n + 1$' is not a statement (it's a number, the value of which depends on the value of n).

Some statements, like the last one in Example 40, depend on an integer n. They may be true for some values of n and false for other values of n. Can we possibly hope to prove whether a statement holds true for an infinite number of integers? Well, let's have a look:

Example 41 Consider the statement

$$2^n > n^2.$$

When $n = 0$, then

$$2^n = 2^0 = 1 > 0^2 = n^2,$$

so the statement is true when $n = 0$. On the other hand, when $n = 2$, we have

$$2^n = 2^2 = 4 = 2^2 = n^2$$

so the statement is not true when $n = 2$. We can check whether the statement is true for the first few values of n:

n	2^n	n^2	'$2^n > n^2$'
0	1	0	True
1	2	1	True
2	4	4	False
3	8	9	False
4	16	16	False
5	32	25	True
6	64	36	True
7	128	49	True
8	256	64	True

After studying this table and thinking about 2^n and n^2, we might guess that $2^n > n^2$ for all integers $n \geq 5$. The evidence looks even stronger once we compare the graphs of 2^x and x^2 (Figure 5.1).

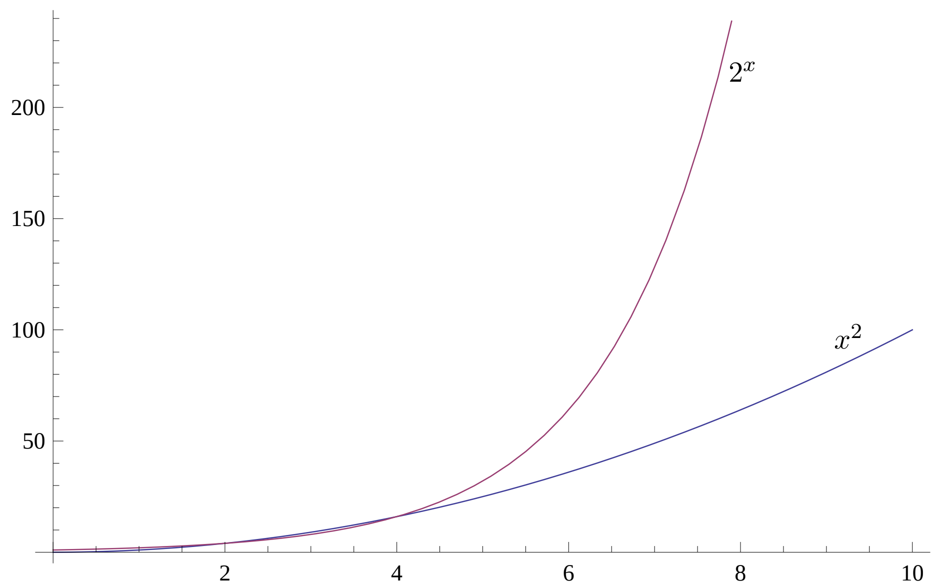


Figure 5.1: 2^x versus x^2.

Although the evidence suggests that

'$2^n > n^2$ for all integers $n \geq 5$',

this by itself is not a *proof*. Since this is MAM1000W, we would like to *prove* that it's true, or *prove* that it's not true.

We know that when $n = 8$, the statement is true, i.e., that $2^n > n^2$. Can we use this to prove that the statement is true for the next integer, $n + 1 = 9$? Well:

$$\begin{aligned} 2^{n+1} &= 2 \cdot 2^n \\ &> 2 \cdot n^2 \qquad \text{(since, by assumption, } 2^n > n^2.\text{)} \\ &> \left(1 + \frac{1}{n}\right)^2 n^2 \qquad \text{(since } 2 > (1 + 1/n)^2 \text{ for } n \geq 5\text{)} \\ &= (n+1)^2 \end{aligned}$$

So, when $n = 8$, we have $2^{n+1} > (n + 1)^2$. This proves that the statement is true when $n = 9$. But then we can apply the same calculations as before to the case when $n = 9$ to prove that the statement is true when $n = 10$. And then, since it's true for $n = 10$, the same argument proves that it's true for $n = 11$, which implies that the statement is true for $n = 12$, which in turn implies that the statement is true for $n = 13$, and so on. It seems clear, therefore, that the statement is true for every integer $n \geq 5$.

With the benefit of hindsight, there was no reason to compare the values of 2^n and n^2 for $n \leq 8$: Since (i) the statement is true when $n = 5$, and, (ii) the calculation that we did is also true for $n \geq 5$, we could have concluded immediately from (i) and (ii) that the statement is true for all $n \geq 5$.

5.2 The Principle of Mathematical Induction

In Example 41, our proof that '$2^n > n^2$ for all integers $n \geq 5$' looked like this:

1. We proved that when $n = 5$, we have $2^n > n^2$, and,
2. We proved that if the statement is true for some integer $n \geq 5$, then it must be true for the integer $n + 1$.

This is an example of a proof by *mathematical induction* (which we call *induction* for short). Mathematical induction is a common proof technique in many areas of mathematics. The reason that a proof of this kind works is because of the *Principle of Mathematical Induction*:

Principle of Mathematical Induction (PMI)
Suppose that
1. A statement S is true when $n = k$ (where k is some specific integer), and,
2. For all $n \geq k$, whenever S is true for some integer n, then it must also be true for the integer $n + 1$.

Then S is true for every integer $n \geq k$.

In general, every inductive proof consists of the two steps above:

1. We first prove that the statement is true when $n = k$ (the *Base Case*). In Example 41, we had $k = 5$.
2. We then prove that if the statement is true for some integer $n \geq k$ (the *Inductive Hypothesis*), then it must be true for the integer $n + 1$. This is called the *Inductive Step.*

Example 42 We shall use induction to prove that for all positive integers n,

$$1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n + 1),$$

which we can write using sigma notation as

$$\sum_{i=1}^{n} i = \frac{1}{2}n(n+1).$$

Base Case: When $n = 1$, we have $\sum_{i=1}^{1} i = 1 = \frac{1}{2}1(1+1)$. So the statement is true when $n = 1$ and the Base Case is proved.

Inductive Step: Suppose that the statement is true for some integer $n \geq 1$. Then we know that

$$\sum_{i=1}^{n} i = \frac{1}{2}n(n+1)$$

(this is the Inductive Hypothesis). We must use this to prove that the statement is true for the integer $n+1$. We proceed as follows:

$$\begin{aligned}
\sum_{i=1}^{n+1} i &= \sum_{i=1}^{n} i + (n+1) \\
&= \frac{1}{2}n(n+1) + (n+1) \qquad \text{(by the Inductive Hypothesis)} \\
&= (n+1)(\frac{n}{2}+1) \\
&= \frac{1}{2}(n+1)(n+2)
\end{aligned}$$

This means the statement is true for the integer $n+1$.

We have proved that the statement is true for $n = 1$ (the Base Case) and we have proved that whenever the statement is true for an integer n, it is also true for the integer $n+1$ (the Inductive Step). Hence, by the Principle of Mathematical Induction, the statement is true for all positive integers n.

Example 43 We shall use induction to prove that if $x \in (-1, \infty)$ and n is a nonnegative integer, then[1]

$$(1+x)^n \geq 1 + nx.$$

Base Case: When $n = 0$, we have $(1+x)^n = (1+x)^0 = 1 = 1 + 0x = 1 + nx$, so we have proved the Base Case.

Inductive Step: Suppose that the statement is true for some integer $n \geq 0$. Then we know that

$$(1+x)^n \geq 1 + nx$$

(this is the Inductive Hypothesis). We must use this to prove that the statement is true for the integer $n+1$. We proceed as follows:

$$\begin{aligned}
(1+x)^{n+1} &= (1+x)^n(1+x) \\
&\geq (1+nx)(1+x) \qquad \text{(by the Inductive Hypothesis)} \\
&= 1 + nx + x + nx^2 \\
&= 1 + (n+1)x + nx^2 \\
&\geq 1 + (n+1)x \qquad \text{(since } nx^2 \geq 0\text{)}
\end{aligned}$$

[1] This is *Bernoulli's Inequality.*

We have proved that the statement is true for $n = 0$ (the Base Case) and we have proved that whenever the statement is true for an integer n, it is also true for the integer $n+1$ (the Inductive Step)[2]. Hence, by the Principle of Mathematical Induction, the statement is true for all nonnegative integers n.

Example 44 We shall use induction to prove that for every positive integer n,

$$8^n - 3^n$$

is divisible by 5.

Base Case: When $n = 1$, $8^n - 3^n = 8^1 - 3^1 = 5$, which is clearly divisible by 5. Hence we have proved the Base Case.

Inductive Step: Suppose that the statement is true for some integer $n \geq 1$. Then there is some positive integer k such that

$$8^n - 3^n = 5k$$

(this is the Inductive Hypothesis). We must use this to prove that the statement is true for the integer $n+1$. We proceed as follows:

$$\begin{aligned} 8^{n+1} - 3^{n+1} &= 8 \cdot 8^n - 3 \cdot 3^n \\ &= 8 \cdot 8^n - 3 \cdot (8^n - 5k) \qquad \text{(by the Inductive Hypothesis)} \\ &= (8-3) \cdot 8^n + 15k \\ &= 5(8^n + 3k) \end{aligned}$$

Since n and k are positive integers, the quantity $8^n + 3k$ is an integer. Hence, $8^{n+1} - 3^{n+1}$ is divisible by 5.

We have proved that the statement is true for $n = 1$ (the Base Case) and we have proved that whenever the statement is true for an integer n, it is also true for the integer $n+1$ (the Inductive Step). Hence, by the Principle of Mathematical Induction, the statement is true for all nonnegative integers n.

Check out some Kahn academy links on induction here (https://www.khanacademy.org/tag/proof-by-induction-math).

[2]Where in the proof did we use the assumption that $x > -1$?

6 Proof by contradiction

In this section, we explore another proof technique: *Proof by contradiction*. In a proof by contradiction, we assume that what we're trying to prove is actually *false*, and then show that this leads to a contradiction. In this section, we give two examples of proof by contradiction. Both of them require some knowledge of prime numbers, and so that is where we begin.

If an integer n is divisible by an integer b, then we say that b is a *factor* or *divisor* of n, and that b *divides* n, written $b|n$.

Example 45 The positive factors of the number 12 are $1, 2, 3, 4, 6$, and 12. Thus, 3 divides 12, which we can write as $3|12$.

An integer $n \geq 2$ is called *prime* if it has exactly two positive factors (1 and itself) or *composite* if it has more than two positive factors. Every integer $n \geq 2$ is either prime or composite.

Example 46

- The number 1 is neither prime nor composite.
- The number 2 is prime since it has exactly two positive factors: 1 and 2.
- Similarly, the number 3 is prime.
- The number 4 is composite, since it has three positive factors: 1, 2, and 4.
- The number 5 is prime.
- The number 6 is composite, since it has four positive factors: 1, 2, 3, and 6.

Every integer $n \geq 2$ can be written as the product of (one or more) prime numbers (its 'prime factorization').

Example 47 Below are the prime factorizations of some numbers:

- $6 = 2 \cdot 3$
- $10 = 2 \cdot 5$
- $11 = 11$ (the number 11 is prime!)
- $20 = 2 \cdot 2 \cdot 5 = 2^2 \cdot 5$
- $33 = 3 \cdot 11$
- $121 = 11^2$
- $45375 = 3 \cdot 5^3 \cdot 11^2$

If the factors don't have to be prime numbers, then integers can be factored in more than one way.

Example 48 The number 36 can be written as $36 = 9 \cdot 4$ or $36 = 12 \cdot 3$ or $36 = 6 \cdot 6$ or $36 = 36 \cdot 1$ or $36 = 6 \cdot 1 \cdot 2 \cdot 1 \cdot 3 \cdot 1$ (and in other ways as well).

However, the *Fundamental Theorem of Arithmetic*[1] says that every integer $n \geq 2$ has a *unique* prime factorization. In mathematics, the word 'unique' means 'exactly one'. So when we say 'Every integer $n \geq 2$ has a *unique* prime factorization', we mean 'Every integer $n \geq 2$ has *exactly one* prime factorization', i.e., every integer $n \geq 2$ can be written as the product of prime factors in exactly one way

[1] The proof of which you will see if you continue with mathematics beyond 1st year.

(ignoring the different orders in which we can write the prime factors down).

Example 49 The number 36 has the unique prime factorization

$$36 = 2 \cdot 2 \cdot 3 \cdot 3 = 2^2 \cdot 3^2.$$

We can write its prime factors down in a different order, like

$$36 = 2 \cdot 3^2 \cdot 2,$$

but they're still the same prime factors (two 2s and two 3s).

Finding the prime factors of a positive integer is easy when the integer is small, but can be difficult when the integer is large (say, 300 digits). The difficulty of this problem is one of the fundamental building blocks used in *cryptography*: the branch of mathematics that concerns itself with secret messages.

We now give our first example of proof by contradiction.

Theorem 50. *There are infinitely many prime numbers.*

Proof. We shall assume that what we want to prove is actually *false* — i.e., we shall assume that there are *not* infinitely many prime numbers — and show that, as a consequence, something impossible happens. Suppose then, by way of contradiction, that there are finitely many prime numbers. This is the same as saying that there are exactly k prime numbers, where k is some positive integer. Call the prime numbers $p_1, p_2, \ldots, p_k$, and consider the integer

$$n = p_1 p_2 \cdots p_k + 1.$$

Certainly, $n \geq 2$, and thus n has a prime factorization. It is thus divisible by at least one of the prime numbers $p_1, p_2, \ldots, p_k$. But $n - 1 = p_1 p_2 \cdots p_k$ is exactly divisible by every prime number p_i, which means that n is not divisible by any of them! This is a contradiction. Hence, the assumption that there are finitely many prime numbers is false, and so there must be infinitely many prime numbers. □

Our second example of proof by contradiction uses prime numbers as well. First, though, we need some more definitions. A real number n is called *rational* if there are integers p and q, with $q \neq 0$, such that $n = p/q$. A real number that is not rational is *irrational*.

Example 51

- The number $\frac{2}{3}$ is clearly rational.
- So is $-\frac{2}{3}$, which can be written as $\frac{-2}{3}$ or $\frac{2}{-3}$ (these are not the only possibilities).
- The number 4 is rational, as $4 = \frac{12}{3} = \frac{-40}{-10} = \frac{100}{25} = \cdots$.
- So is 0, because $0 = \frac{0}{1} = \frac{0}{2} = \frac{0}{3} = \cdots$.

With a little thought, it's easy to see that all the integers and all the fractions (both proper and mixed) are rational. To find an irrational number, though, is not so easy. To prove that a number n is rational, all we have to do is find two integers p and q such that $n = p/q$. To prove that n is irrational, however, we have to prove that it is *impossible* to find such a pair of numbers. This is where proof by contradiction is useful, as we shall shortly see. We make two important observations.

1. If n is rational, then there are infinitely many different ways we can choose the integers p and q. For example, if $n = \frac{2}{3}$, then we can write $n = \frac{4}{6} = \frac{6}{9} = \frac{8}{12} \cdots$. In each case, the two integers p and q have a *common factor*, e.g., $\frac{4}{6} = \frac{2 \cdot 2}{2 \cdot 3}$ (there is a common factor of 2) and $\frac{6}{9} = \frac{3 \cdot 2}{3 \cdot 3}$ (a common factor of 3), which may be divided out to give a pair p, q of integers with no common factor. This will play a pivotal role in the proof of Theorem 53.
2. The second observation is the following:

   Lemma 52. *Let p be a prime number and n a positive integer. Then n is divisible by p if and only if n^2 is divisible by p.*

   Proof. By the Fundamental Theorem of Arithmetic, n has a unique prime factorization

   $$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k},$$

   where each p_i is a different prime number and each e_i is a positive integer. Thus,

   $$\begin{aligned} n^2 &= (p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k})^2 \\ &= (p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k})(p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}) \\ &= p_1^{e_1+e_1} p_2^{e_2+e_2} \cdots p_k^{e_k+e_k} \\ &= p_1^{2e_1} p_2^{2e_2} \cdots p_k^{2e_k}. \end{aligned}$$

   By the Fundamental Theorem of Arithmetic, this must be the prime factorization of n^2. Hence, the integers n and n^2 have exactly the same prime factors, and so n is divisible by p if and only if n^2 is divisible by p. □

We are now ready to see our second example of proof by contradiction:

Theorem 53. *$\sqrt{2}$ is irrational.*

Proof. Once again, we shall assume that what we want to prove is actually *false* and show that, when we do, something impossible happens.

Assume, to the contrary, that $\sqrt{2}$ is rational. Then there are integers m and n, with $n \neq 0$, such that

$$\sqrt{2} = \frac{m}{n} \tag{6.1}$$

and such that (as per our previous discussion) m and n have no common factor. Squaring both sides of (6.1), we obtain

$$2 = \frac{m^2}{n^2},$$

which implies that

$$m^2 = 2n^2. \tag{6.2}$$

Hence, m^2 is an even integer. Thus, by Lemma 52, m is an even integer, and so there is some integer k such that $m = 2k$. Substituting into (6.2), we get $4k^2 = 2n^2$, and hence $2k^2 = n^2$. Thus, n^2 is an even integer, which implies by Lemma 52 that n is an even integer. But then m and n are both multiples of 2, which means they have a common factor. This contradicts our choice of m and n. Hence, no such integers m and n exist, and $\sqrt{2}$ is irrational. □

Some proofs that are not in Stewart 7

Theorem 54 (The Product Rule)**.** *If f and g are differentiable, then*

$$(fg)' = f'g + fg'.$$

Proof. From the definition of the derivative,

$$\begin{aligned}
(fg)'(x) &= \lim_{h\to 0} \frac{(fg)(x+h) - (fg)(x)}{h} \\
&= \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\
&= \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\
&= \lim_{h\to 0} \frac{[f(x+h) - f(x)]g(x+h) + [g(x+h) - g(x)]f(x)}{h} \\
&= \lim_{h\to 0} \left[\frac{f(x+h) - f(x)}{h} g(x+h) + \frac{g(x+h) - g(x)}{h} f(x)\right]
\end{aligned}$$

Now:

- Since f and g are differentiable, $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ and $\lim_{h\to 0} \frac{g(x+h)-g(x)}{h}$ exist (and, of course, $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$ and $\lim_{h\to 0} \frac{g(x+h)-g(x)}{h} = g'(x)$).
- Since $f(x)$ has no dependence on h, the limit $\lim_{h\to 0} f(x)$ clearly exists (for the purpose of evaluating this limit, we can treat $f(x)$ as a constant). Thus $\lim_{h\to 0} f(x) = f(x)$.
- Since g is differentiable, by Theorem 4 in Stewart 2.7, g is continuous. Hence, by Theorem 8 in Stewart 2.4,
$$\lim_{h\to 0} g(x+h) = g(\lim_{h\to 0}(x+h)) = g(x).$$

Since all four of these limits exist, by the Limit Laws, we can continue our calculation as follows:

$$\begin{aligned}
(fg)'(x) &= \lim_{h\to 0} \left[\frac{f(x+h) - f(x)}{h} g(x+h) + \frac{g(x+h) - g(x)}{h} f(x)\right] \\
&= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \cdot \lim_{h\to 0} g(x+h) + \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} \cdot \lim_{h\to 0} f(x) \\
&= f'(x)g(x) + f(x)g'(x)
\end{aligned}$$

□

Theorem 55 (Rolle's Theorem)**.** *Suppose that f is continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$, then there is a number $c \in (a, b)$ where $f'(c) = 0$.*

Proof. We consider two cases:

Case 1: f is constant on $[a, b]$. Then $f'(x) = 0$ for all $x \in (a, b)$, so we can take c to be any number in (a, b).

Case 2: f is not constant on $[a, b]$. We consider two subcases:

Case 2.1: There is some $d \in (a, b)$ where $f(d) > f(a)$. Since f is continuous on $[a, b]$, by the Extreme Value Theorem, f has an absolute maximum at some $c \in [a, b]$. Since $f(c) \geq f(d) > f(a)$, we can't have $c = a$ or $c = b$, so we must have $c \in (a, b)$. Thus $f'(c)$ exists, and hence, by Fermat's Theorem, $f'(c) = 0$.

Case 2.2: There is some $d \in (a, b)$ where $f(d) < f(a)$. Since f is continuous on $[a, b]$, by the Extreme Value Theorem, f has an absolute minimum at some $c \in [a, b]$. Since $f(c) \leq f(d) < f(a)$, we must have $c \in (a, b)$. Thus $f'(c)$ exists, and hence, by Fermat's Theorem, $f'(c) = 0$.

□

Theorem 56 (Mean Value Theorem)**.** *If f is continuous on $[a, b]$ and differentiable on (a, b), then there is a number $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Consider the function

$$h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Since $f(x)$ and $x - a$ are continuous on $[a, b]$, the function $h(x)$ is continuous on $[a, b]$. Similarly, since $f(x)$ and $x - a$ are differentiable on (a, b), the function $h(x)$ is differentiable on (a, b). Notice also that

$$h(a) = f(a) - f(a) + \frac{f(b) - f(a)}{(b - a)}(a - a) = 0$$

and

$$h(b) = f(b) - f(a) - \frac{f(b) - f(a)}{b - a}(b - a) = 0.$$

To summarize: The function $h(x)$ is continuous on $[a, b]$, differentiable on (a, b), and $h(a) = h(b)$. Therefore, by Rolle's Theorem, there is some $c \in (a, b)$ where $h'(c) = 0$. Now,

$$h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

which means that

$$h'(c) = 0 = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

Thus,

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

□

Theorem 57. *If $f'(x) = 0$ for all $x \in (a, b)$, then $f(x)$ is constant on (a, b).*

Proof. We shall use proof by contradiction. Suppose, to the contrary, that f is a function with $f'(x) = 0$ for all $x \in (a, b)$ but that f is not constant on (a, b). Since f is not constant on (a, b), we can find a_1, b_1 in (a, b) where $a_1 < b_1$ and $f(a_1) \neq f(b_1)$. Since $f'(x) = 0$ for all $x \in (a, b)$, the function f is certainly differentiable on (a, b), which implies that it's differentiable on (a_1, b_1) and hence (by Theorem 4 in Stewart 2.7) continuous on $[a_1, b_1]$. Thus, by the Mean Value Theorem, there is some $c \in (a_1, b_1)$ where

$$f'(c) = \frac{f(b_1) - f(a_1)}{b_1 - a_1}.$$

But $f(a_1) \neq f(b_1)$, so $f'(c) \neq 0$. This is a contradiction, and hence f must be constant on (a, b). □

Theorem 58 (Mean Value Theorem for Integrals)**.** *If f is continuous on $[a, b]$ (where $a < b$), then there exists $c \in (a, b)$ such that*

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\ dx.$$

Proof. Let $F(x) = \displaystyle\int_a^x f(t)\ dt$. From the Fundamental Theorem of Calculus, F is continuous on $[a, b]$ and differentiable on (a, b). Hence, by the Mean Value Theorem (for derivatives!), there exists $c \in (a, b)$ such that

$$F'(c) = \frac{F(b) - F(a)}{b - a}.$$

But $F'(c) = f(c)$, so

$$f(c) = F'(c) = \frac{F(b) - F(a)}{b - a} = \frac{\int_a^b f(t)\ dt - \int_a^a f(t)\ dt}{b - a} = \frac{1}{b-a} \int_a^b f(t)\ dt.$$

Since $\displaystyle\int_a^b f(t)\ dt = \int_a^b f(x)\ dx$, this finishes the proof. □

An alternative proof: Since f is continuous on $[a, b]$, by the Extreme Value Theorem, f has a global maximum and a global minimum on $[a, b]$, i.e., there exist $x_1, x_2 \in [a, b]$ such that $f(x_1) \leq f(x) \leq f(x_2)$ for all $x \in [a, b]$. Therefore, by the Comparison Properties for Integrals,

$$f(x_1)(b - a) \leq \int_a^b f(x)\ dx \leq f(x_2)(b - a)$$

and thus

$$f(x_1) \leq \frac{1}{b-a} \int_a^b f(x)\ dx \leq f(x_2).$$

Therefore, by the Intermediate Value Theorem, there is a number c between x_1 and x_2 where

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\ dx.$$

□